3LPo1G-05 Assessing Vanadium Oxide Smart Insulation for Fusion-relevant HTS Magnets

Master's Thesis at MIT to graduate from TUM in Aerospace Engineering (March - November 2026)

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Summary

Proxima Fusion presents a **novel quench protection concept [1]** that

- **Circumvents high voltages** during a quench/fast discharge scenario
- Promises a more **uniform energy deposition due to >10m/s normal zone propagation velocities** (NZPV) on stack level

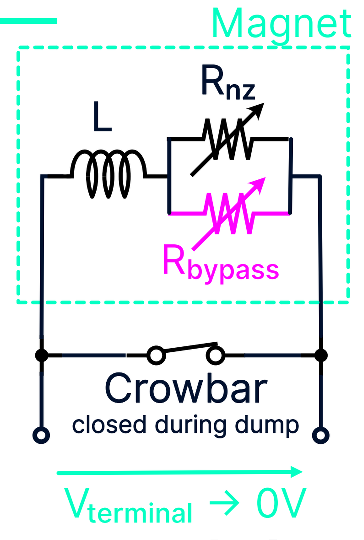
Simulation suggests normal zone propagation velocities of >10m/s is achievable with a stack of 5 ReBCO tapes. An experiment is being built to study this effect.

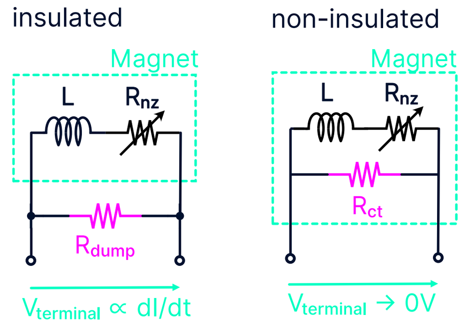
1 No terminal voltage during quench

$$L\frac{\mathrm{d}I}{\mathrm{d}t} = -R_{\mathrm{nz}}(t)I, \quad V_{\mathrm{term}} = 0$$

No terminal voltage - inductive voltage equals voltage drop over conductor, independent of the stored energy.

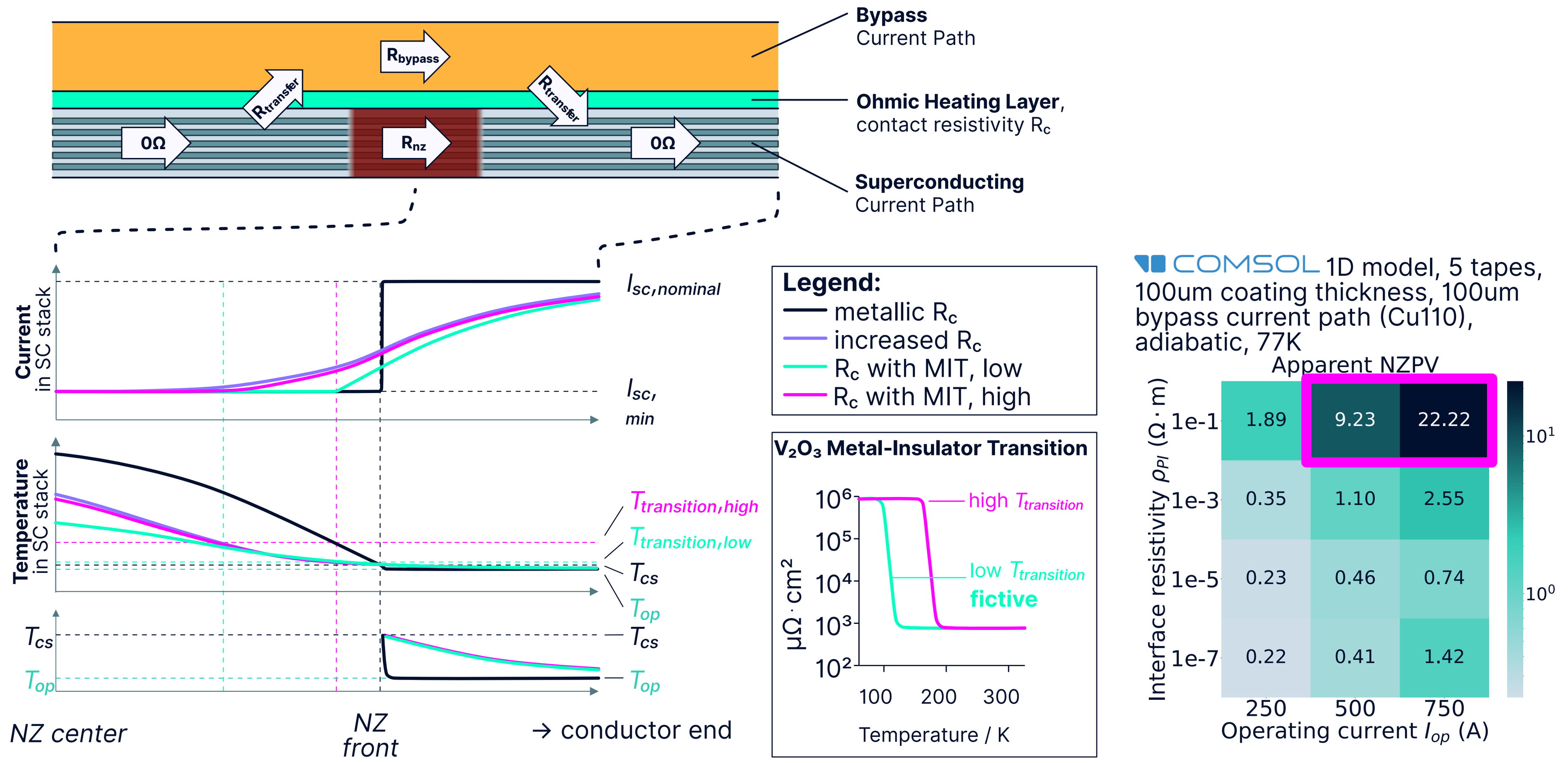
$$Al_{\mathrm{nz}}\int_{T_{\mathrm{op}}}^{T_{\mathrm{safe}}} C(T)\,\mathrm{d}T \geq \frac{1}{2}LI^2$$





Total **magnetic energy is transferred to** conductor **enthalpy**. Only possible if the turn quenches uniformly. **This requires fast NZPV.**

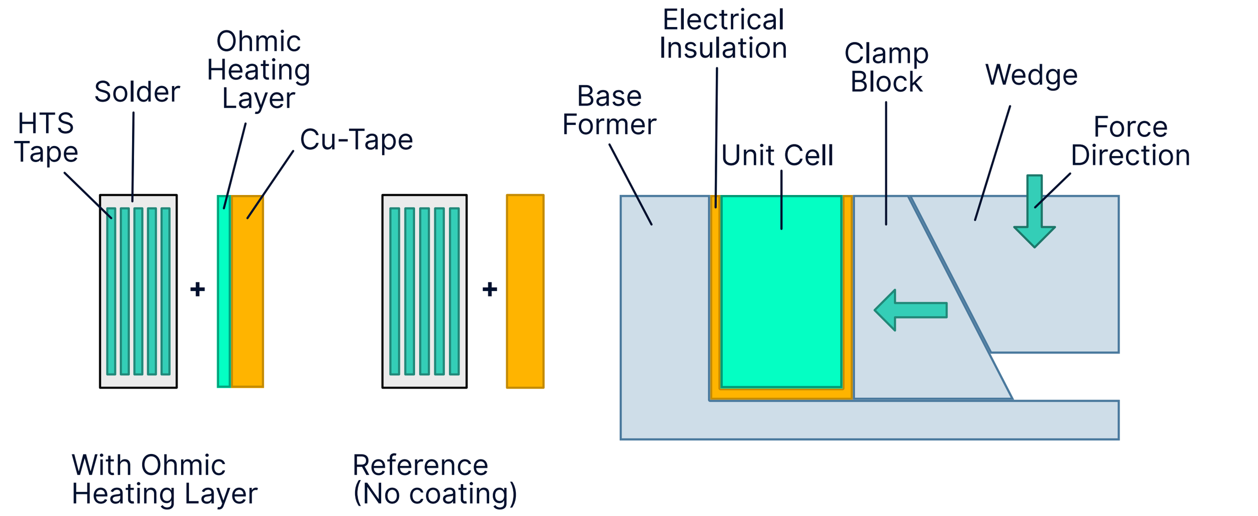
2 Normal zone propagation velocity > 10 m/s



3 Test with 1d physics instead of a coil

Experiment setup

- Contact pressure from same direction relative to tape
- Quasi-adiabatic
- $T_{peak} < T_{safe} = 400K$

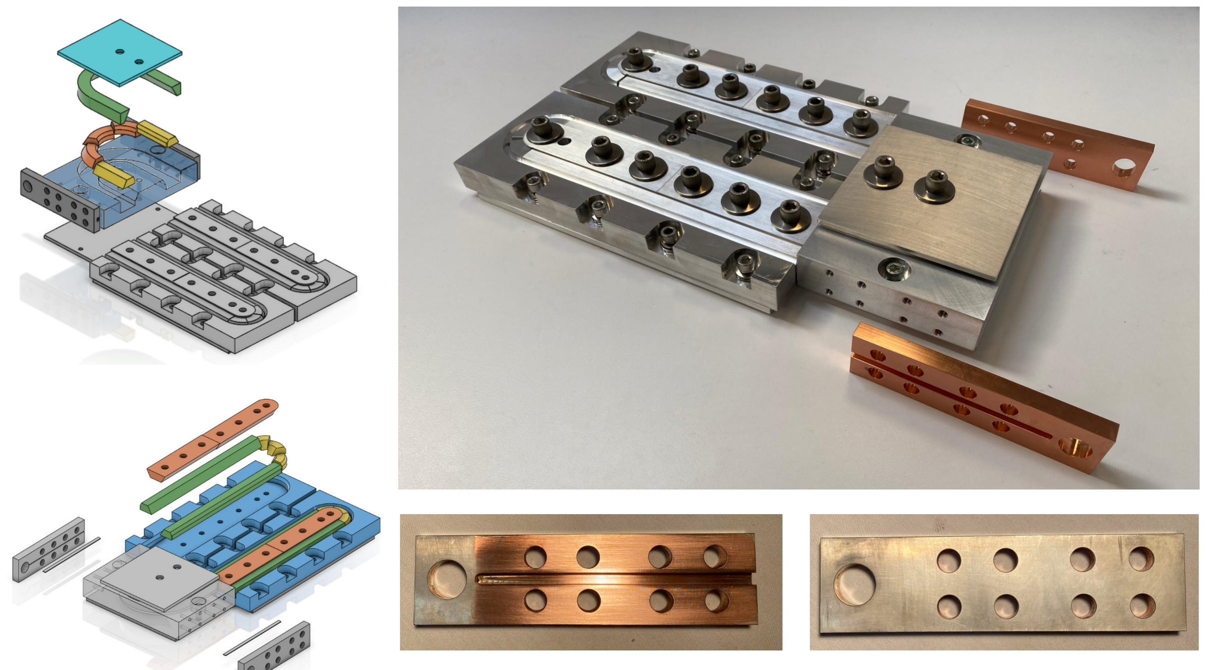


Section view of fast NZPV conductor & reference

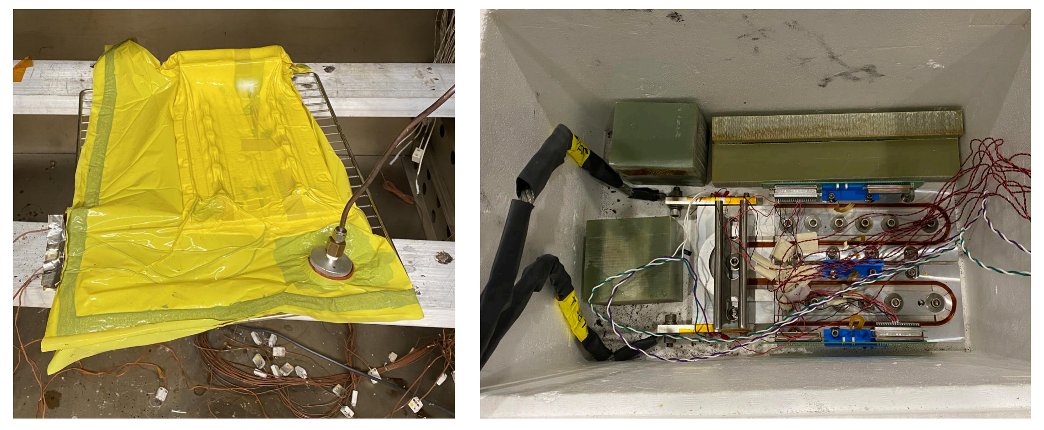
- Materials: Alu 6061 & Cu110
- Belleville washers for sustained pressure during temperature cycles
- Torque deduced from FUJI film pressure distribution

Cold bus terminals

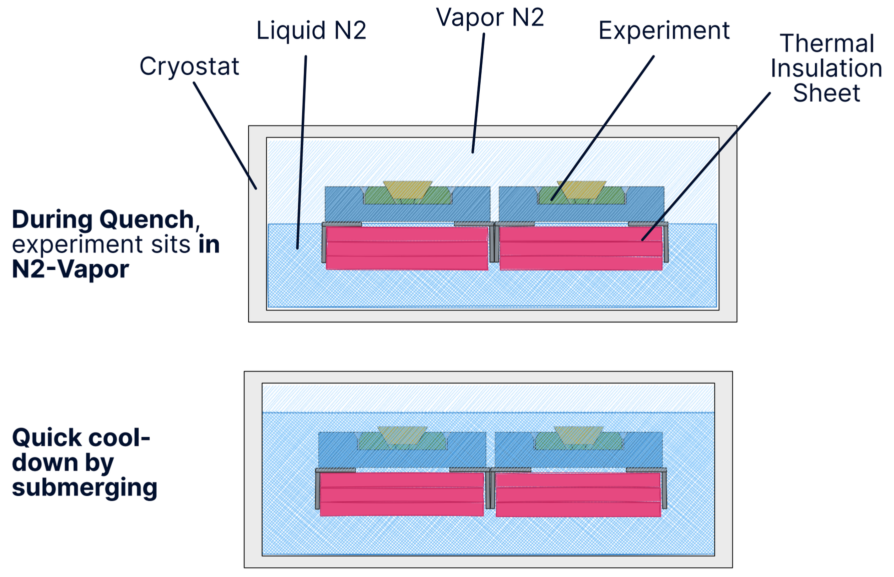
- 40 HTS tapes each, precise current injection into conductor
- Silver-plated for lowest current transfer length



Assembly & cold bus

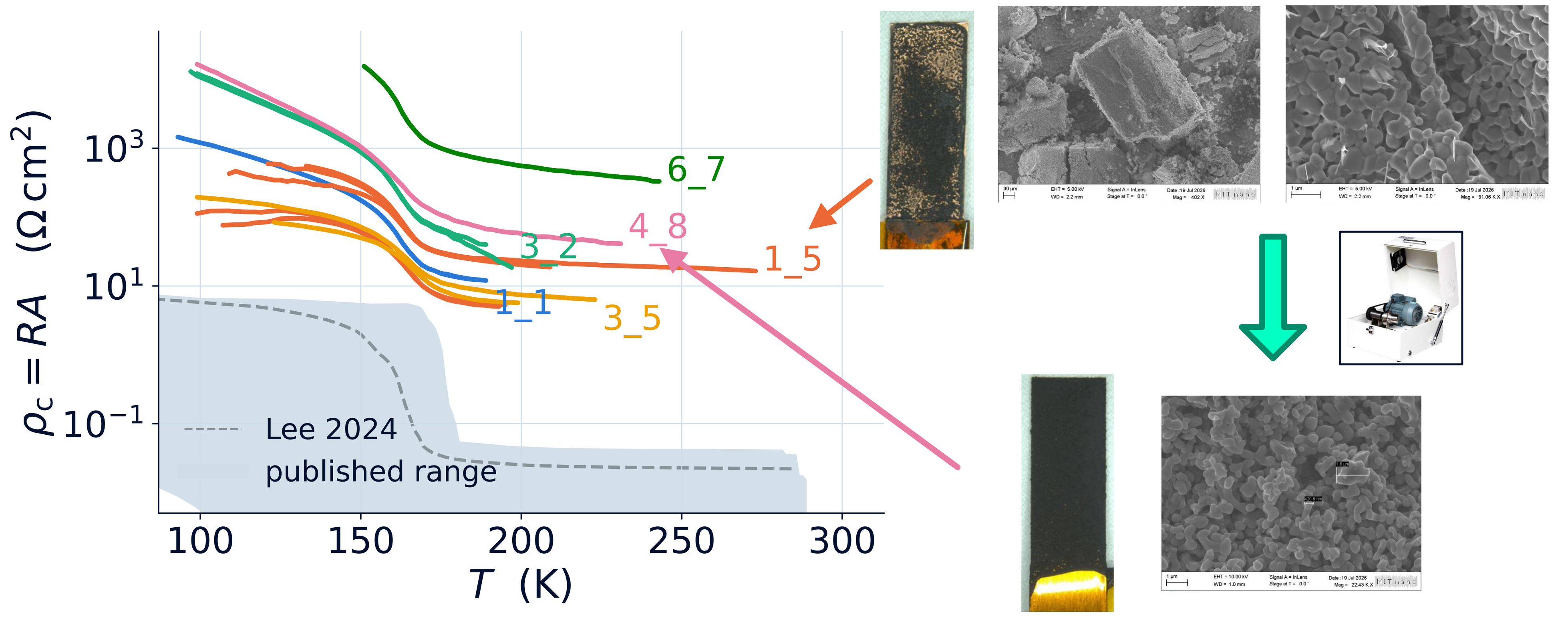


Re-flow solder process & instrumented assembly
same former can be used for soldering and testing



Experiment operations
fast cool-down by dunking into LN2 to allow for low downtime (<2 min) between quenches

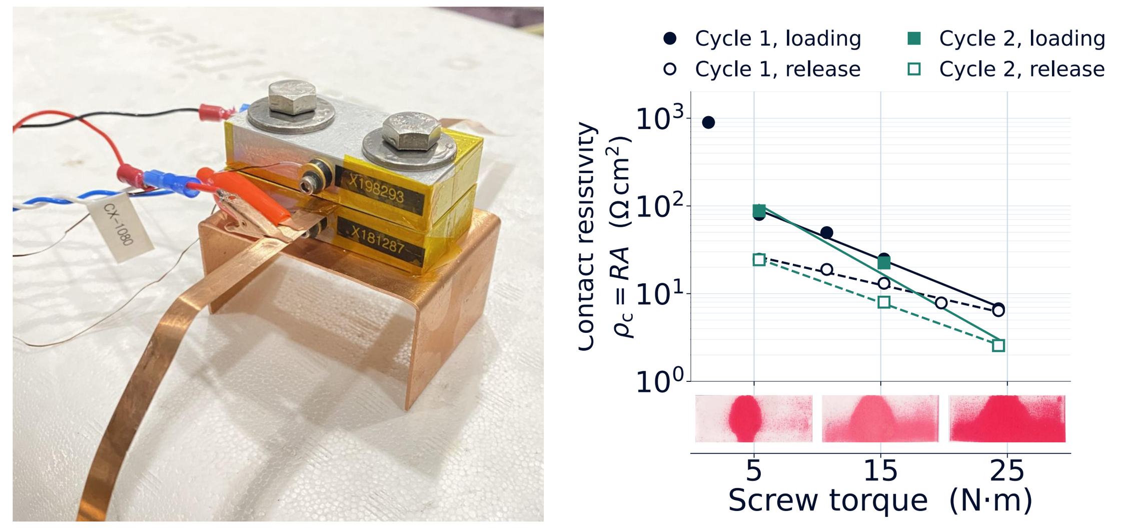
4 Too high coating resistivity increases experiment complexity



Coupon tests of coating for prototyping
Lower coating resistance decreases minimum conductor length. References [4-6]

Tuning resistivity is a challenge

- Currently achievable coating resistivity too high. Sample length of >40m.
- Coating thickness cannot regulate resistivity enough, but improves hiding power
- Doping V_2O_3 powder with less resistive material not yet successful
- Pressure regulates resistivity



Contact Pressure regulates resistance reliably
Coating resistance can be adjusted by 1 OOM

Acknowledgement

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References

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